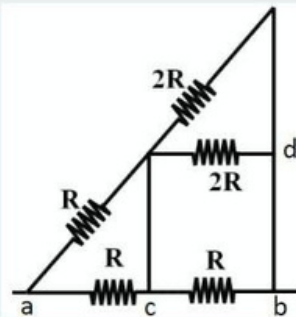


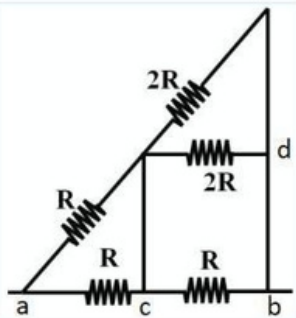
2



In this resistors circuit $V_{ab} = 160V$ and $R = 20 \text{ Ohm}$. What is the equivalent resistor of the circuit? (4 Puan)

- ☐ 120/11 Ohm
- ☐ 40 Ohm
- ☐ 40/3 Ohm
- ☐ 20 Ohm
- ☐ 10 Ohm

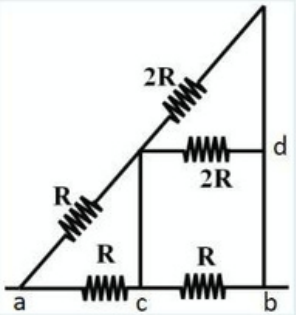
3



In this resistors circuit $V_{ab} = 160V$ and $R = 20 \text{ Ohm}$. What is the total power of this circuit? (4 Puan)

- ☐ 2349 W
- ☐ 2560 W
- ☐ 3200 W
- ☐ 1280 W
- ☐ 5120 W

4

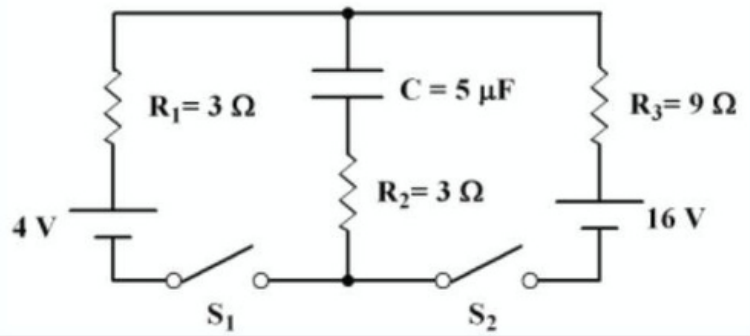


In this resistors circuit $V_{ab} = 160V$ and $R = 20 \text{ Ohm}$. What is the current on R between cb points? (4 Puan)

- ☐ 2 A
- ☐ 1.6 A
- ☐ 4 A
- ☐ 8 A
- ☐ 1 A

5

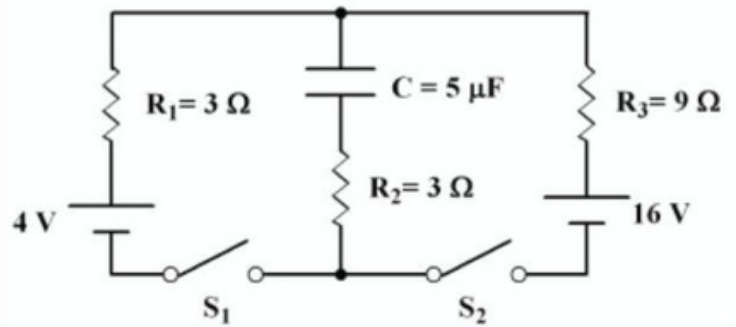
A circuit given in the figure is used for charging the $5 \mu F$ capacitor. What is the absolute value of the current on the $4V$ battery long after both of the switches (S_1 and S_2) are closed? (4 Puan)



- ☐ 1 A
- ☐ 0.5 A
- ☐ 10/3 A
- ☐ 10/6 A
- ☐ 2 A

6

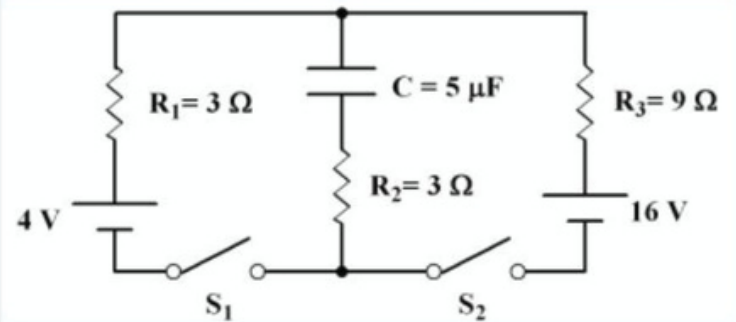
A circuit given in the figure is used for charging the $5 \mu F$ capacitor. What is the charge on the capacitor long after the switches are closed? (4 Puan)



- ☐ $25 \mu C$
- ☐ $35 \mu C$
- ☐ $40 \mu C$
- ☐ $55 \mu C$
- ☐ $90 \mu C$

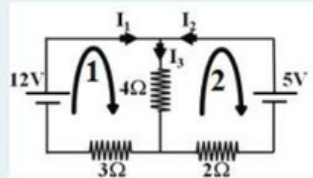
7

A circuit given in the figure is used for charging the $5 \mu F$ capacitor. S_2 switch will be OPEN. What is the time constant of the circuit if only S_1 switch is closing? (4 Puan)



- ☐ $7.5 \mu s$
- ☐ $30 \mu s$
- ☐ $6 \mu s$
- ☐ $15 \mu s$
- ☐ $75 \mu s$

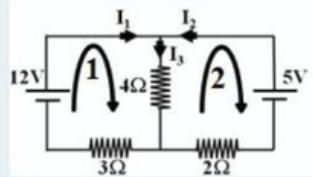
8



In the circuit given in the figure what is the equation from the 2nd loop? (4 Puan)

- ☐ $-5 + 2I_2 + 4I_3 = 0$
- ☐ $5 + 2I_2 + 4I_3 = 0$
- ☐ $-5 - 2I_2 + 4I_3 = 0$
- ☐ $-5 + 2I_2 - 4I_3 = 0$
- ☐ $5 + 4I_2 + 2I_3 = 0$

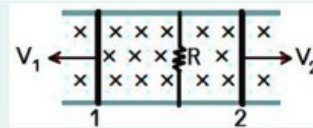
9



In the circuit given in the figure what are values of current $I_1 : I_2 : I_3$ (4 Puan)

- ☐ $2A : -0.5A : 1.5A$
- ☐ $2A : 0.5A : 1.5A$
- ☐ $1.5A : 0.5A : 2A$
- ☐ $-2A : 0.5A : -1.5A$
- ☐ $1.5A : 0.5A : 2A$

10

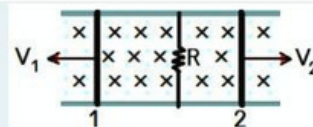


There are two parallel metal bars where a resistance R is connected to both of them. There are also two movable wires, the 1st one moves towards left with speed V_1 and the 2nd one moves towards right with speed V_2 . Length of movable wires are L . There is also a constant magnetic field B into the page. There is NO resistance on the system except resistance R and there is NO frictional force in the system. What are the magnetic flux change (manyetik aki degisimi) in the LEFT loop (between 1st wire and the resistance R) and the in RIGHT loop (between the 2nd wire and the resistance R) respectively? LEFT : RIGHT (4 Puan)

$$R = 2\Omega : V_1 = 40m/s : V_2 = 60m/s : L = 50cm : B = 0.2T$$

- ☐ $4V : 6V$
- ☐ $40V : 60V$
- ☐ $2V : 3V$
- ☐ $1V : 1.5V$
- ☐ $20V : 30V$

11

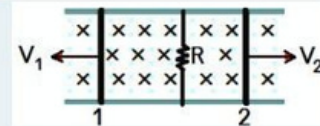


There are two parallel metal bars where a resistance R is connected to both of them. There are also two movable wires, the 1st one moves towards left with speed V_1 and the 2nd one moves towards right with speed V_2 . Length of movable wires are L . There is also a constant magnetic field B into the page. There is NO resistance on the system except resistance R and there is NO frictional force in the system. What is the direction of the INDUCED CURRENT on the 1st and 2nd wire respectively? 1st : 2nd (4 Puan)

$$R = 2\Omega : V_1 = 40m/s : V_2 = 60m/s : L = 50cm : B = 0.2T$$

- ☐ UP : DOWN
- ☐ UP : UP
- ☐ DOWN : UP
- ☐ DOWN : DOWN
- ☐ There is NO Current on each wire !

12

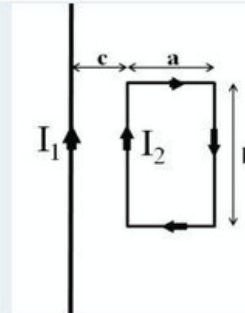


There are two parallel metal bars where a resistance R is connected to both of them. There are also two movable wires of length L , the 1st one moves towards left with speed V_1 and the 2nd one moves towards right with speed V_2 . There is also a constant magnetic field B into the page. There is NO extra resistance on the system except resistance R and there is NO frictional force in the system. What is the direction and the magnitude of the current on resistance R ? (4 Puan)

$$R = 2\Omega : V_1 = 40m/s : V_2 = 60m/s : L = 50cm : B = 0.2T$$

- ☐ UP : 1 A
- ☐ UP : 5 A
- ☐ DOWN : 1 A
- ☐ DOWN : 5 A
- ☐ No Current !

13



There is very long wire carrying a current of I_1 . c distance away there is a rectangular loop which carries a current of I_2 . The dimensions of rectangular loop is a and b .

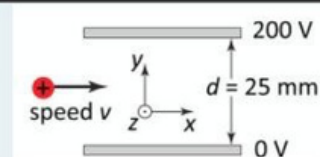
i) find the direction and magnitude forces on each sides of the rectangular loop.

ii) find the total force acting on the loop. Is it zero?

iii) Imagine $I_2 = 0$ and rectangular loop is moving LEFT with a constant velocity v (getting far from long wire)

--> What is the direction of the INDUCED current on rectangular loop ? ---> What is the induced current value if the resistance of the rectangular loop is R ? (Anonim olmayan soru) (28 Puan)

14



A proton enters the region between the two plates shown in the figure with a speed $v = 2000000$ m/s in the x -direction. The potential of the top and bottom plate is 200 V and 0 V, respectively, and their distance is $d = 25$ mm. What is the direction and magnitude of the magnetic field B required between the plates for the proton to continue traveling in a straight line along the x -direction? (4 Puan)

$$V = 2 \times 10^6 \frac{m}{s}$$

- ☐ Magnetic field cannot keep the proton traveling straight.
- ☐ $(2.5 \times 10^{-6} \hat{j})T$
- ☐ $(-0.004 \hat{k})T$
- ☐ $(-2.5 \times 10^{-6} \hat{k})T$
- ☐ $(0.004 \hat{j})T$

15

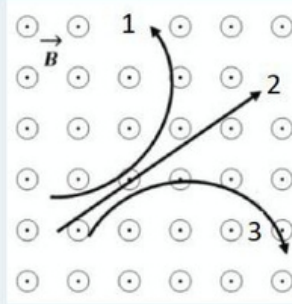
Two negatively charged particles, q_1 and q_2 , are of a distance r apart with $q_1 = 2q_2$. F_{21} is the force q_2 exerts on q_1 and F_{12} is the force q_1 exerts on q_2 . What is the ratio of these two forces F_{12}/F_{21} ?

(4 Puan)

$$\frac{F_{12}}{F_{21}} = ?$$

- ☐ $\frac{1}{2}$
- ☐ 2
- ☐ -1
- ☐ 1
- ☐ -2

16



3 different particle having the same velocity enter constant magnetic field and follow the paths given in the figure. B magnetic field direction is out of the page. What are the electric charge types of these particles in 1 : 2 : 3, respectively.

(4 Puan)

- ☐ Positive : Neutral : Negative
- ☐ Neutral : Positive : Negative
- ☐ Negative : Neutral : Positive
- ☐ More information is needed to answer the question !
- ☐ Negative: Positive: Neutral

17

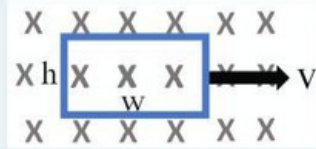
For an electron moving in a direction opposite to the electric field

(4 Puan)

$$U=qV$$

- ☐ its potential energy increases and its electric potential increases.
- ☐ its potential energy increases and its electric potential decreases.
- ☐ its potential energy decreases and its electric potential decreases.
- ☐ both the potential energy and the electric potential remain constant.
- ☐ its potential energy decreases and its electric potential increases.

18



A rectangular metal loop is pulled with a constant velocity V of 5m/s towards right in a constant magnetic field B of 0.5 Tesla which is into the page. The rectangular loop has width $w=4\text{cm}$ and height $h=2\text{cm}$. What is the value for induced EMF and the direction of the induced current ?

(4 Puan)

CW: Clockwise (saat yönü)

CCW: CounterClockWise (saat yönü tersi)

- ☐ 0.4 mV ; CCW
- ☐ 0.4 mV ; CW
- ☐ 0 V ; No current !
- ☐ 2 mV ; CW
- ☐ 2 mV ; CCW

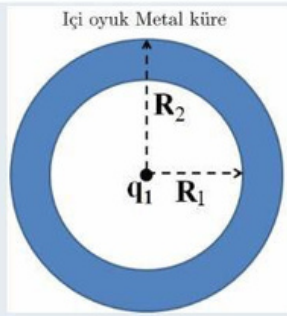
19

If a piece of wire is stretched by 2 times (its final length becomes 2 times of its initial length) what happens to its resistance? (Density of the wire is constant)

(4 Puan)

- ☐ increases 2 times (twice)
- ☐ decreases 2 times (twice)
- ☐ stays constant
- ☐ increases 4 times
- ☐ decreases 4 times

2



In the figure there is a point charge q_1 at the center of the spherical void of a metal sphere. The charge of the metal sphere is unknown. The inner radius is $R_1 = 20\text{ cm}$ and the outer radius is $R_2 = 30\text{ cm}$.

The magnitude of the electric field at 15 cm from the center is given as $E(r = 15\text{ cm}) = 10000\text{ N/C}$ and the direction of this electric field is inward (iç doğru).

The magnitude of the electric field at 30 cm is given as $E(r = 30\text{ cm}) = 10000\text{ N/C}$ and the direction is outwards (dış doğru).

What is q_1 ?

(4 Puan)

$1\text{ nano} = 10^{-9}$

- ☐ -100 nC
- ☐ $+100\text{ nC}$
- ☐ $+25\text{ nC}$
- ☐ -25 nC
- ☐ $+125\text{ nC}$

3

What is the total charge at the inner radius, at $r = 20\text{ cm}$? (iç çeperdeki yük nedir?)

(4 Puan)

$1\text{ nano} = 10^{-9}$

- ☐ $+100\text{ nC}$
- ☐ -100 nC
- ☐ -25 nC
- ☐ $+25\text{ nC}$
- ☐ $+0\text{ nC}$

4

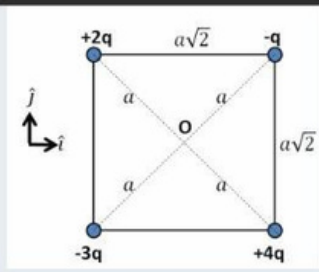
What is the charge at the outer surface of the sphere at $r = 30\text{ cm}$? (En dış yüzeydeki yük nedir?)

(4 Puan)

$1\text{ nano} = 10^{-9}$

- ☐ $+100\text{ nC}$
- ☐ $+25\text{ nC}$
- ☐ -100 nC
- ☐ $+125\text{ nC}$
- ☐ -25 nC

5



In the figure charges have been placed at the corner of a square where the distance from charges to the center O point is a . What is the magnitude and direction of total electric field at the center of the square

(4 Puan)

$$E = \frac{kq}{a^2}$$

- ☐ $-2\sqrt{2}E\hat{j}$
- ☐ $-2E\hat{j}$
- ☐ $-2\sqrt{2}E\hat{i}$
- ☐ $2\sqrt{2}E\hat{i}$
- ☐ $-4E\hat{j}$

6

What is the unit for the expression below?

(4 Puan)

$$\frac{r^2 \epsilon_0}{q}$$

- ☐ $\frac{\text{N}}{\text{C}}$
- ☐ $\frac{\text{m}}{\text{V}}$
- ☐ $\frac{\text{m}^2}{\text{C}}$
- ☐ it is unitless !
- ☐ $\frac{\text{V}}{\text{Nm}}$

7

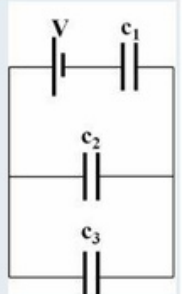
A conducting wire has radius $r = 0.3\text{ mm}$ and is 2.7 m long and it carries a current of 1 A . What 's the electric field in the wire?

(4 Puan)

$$\rho = 3.2 \times 10^{-8} \Omega \text{ m.m}$$

- ☐ $9 \frac{\text{V}}{\text{m}}$
- ☐ $1.1 \frac{\text{V}}{\text{m}}$
- ☐ $\frac{1}{9} \frac{\text{V}}{\text{m}}$
- ☐ $11.1 \frac{\text{V}}{\text{m}}$
- ☐ $0.9 \frac{\text{V}}{\text{m}}$

8



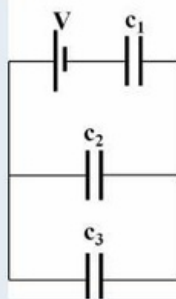
In the circuit capacitors are given as $C_1 = C_2 = C_3 = 10\text{ micro Farad}$ and the battery emf is 10 Volts . What 's the electric potential of C_2 capacitor?

(4 Puan)

$$1\text{ micro Farad} = 10^{-6}\text{ Farad}$$

- ☐ 5 V
- ☐ 3.33 V
- ☐ 6.67 V
- ☐ 10 V
- ☐ 3 V

9

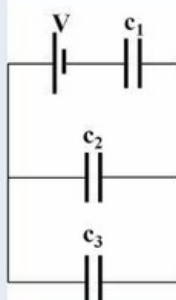


In the circuit capacitors are given as $C_1 = C_2 = C_3 = 10$ micro Farad and the battery emf is 10 Volts. What is the charge of C_1 capacitor?
(4 Puan)

$1 \text{ micro Farad} = 10^{-6} \text{ Farad}$

- ☐ 457 microCoulomb
- ☐ 300 microCoulomb
- ☐ 150 microCoulomb
- ☐ 66.7 microCoulomb
- ☐ 123.4 microCoulomb

10



In the circuit capacitors are given as $C_1 = C_2 = C_3 = 10$ micro Farad and the battery emf is 10 Volts. What is the equivalent capacitance of the circuit?
(4 Puan)

$1 \text{ micro Farad} = 10^{-6} \text{ Farad}$

- ☐ 66.7 microFarad
- ☐ 30 micro Farad
- ☐ 15 microFarad
- ☐ 6.67 microFarad
- ☐ 1.5 microFarad

11

A conducting wire has radius $r = 0.3$ mm and is 2.7 m long and it carries a current of 1 A. What is the electric potential across the wire?
(4 Puan)

$\rho = 3.2 \times 10^{-8} \text{ Ohm.m}$

- ☐ 0.03 V
- ☐ 30 V
- ☐ 3.0 V
- ☐ 0.3 V
- ☐ 0.27 V

12

3.2 A current runs through a copper wire with diameter of 2 mm which is connected to a light bulb. Copper has N number of free electrons per cubic meter. How many electrons pass through the light bulb each second?

(4 Puan)

$N = 8 \times 10^{28} \frac{1}{\text{m}^3} \text{ free electrons}$

- ☐ 0.5×10^{19}
- ☐ 5×10^{19}
- ☐ 6.4×10^{17}
- ☐ 2×10^{17}
- ☐ 2×10^{19}

13

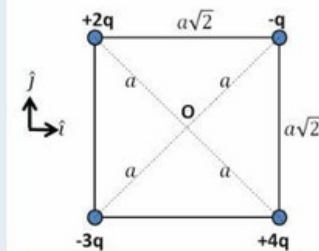
State whether the statements below are True or False for isolated charged parallel plate capacitor.

- I) When the distance between the plates of the capacitor is doubled, the energy stored in the capacitor increases.
- II) Increasing the distance between the plates increases the electric field between the plates.
- III) When the distance between the plates is halved, the charge on the plates stays the same.
- IV) Inserting a dielectric between the plates increases the charge on the plates.
- V) Inserting a dielectric between the plates decreases the energy stored in the capacitor.

(4 Puan)

- ☐ FTFTF
- ☐ TTFTT
- ☐ TFTFF
- ☐ FFTTF
- ☐ TTTFT
- ☐ TFTFT

14



In the figure charges have been placed at the corner of a square where the distance from charges to the center O point is a . What is the total potential energy of the system?
(4 Puan)

$V = \frac{kq}{a}$

- ☐ $\frac{(-11+16\sqrt{2})}{2} qV$
- ☐ $\frac{(11-24\sqrt{2})}{2} qV$
- ☐ $2qV$
- ☐ $\frac{27\sqrt{2}}{2} qV$
- ☐ $\frac{(11-14\sqrt{2})}{2} qV$

15

A uniformly charged insulating sphere has total charge of Q and radius of R .

- Find Electric field Inside and Outside of the sphere using Gauss Law.
- Find Electric potential Inside and Outside of the sphere.
- Draw the graph of potential showing all the important values. (Anonim olmayan soru) 

Charge density is ρ

 Dosyayı karşıya yükle

Dosya sayısı üst sınırı: 1 Tek dosya boyutu üst sınırı: 10MB İzin verilen dosya türleri: Word, Excel, PPT, PDF, Resim, Video, Ses

16

State whether the statements below are True or False about electric field lines.

- Electric field lines point inward toward negative charges.
- Electric field lines make circles around positive charges.
- Electric field lines may cross.
- Electric field lines point outward from positive charges
- A positive point charge released from rest will initially accelerate along the tangent to the electric field line at that point.

(4 Puan)

☐ T F F T T

☐ T F T F T

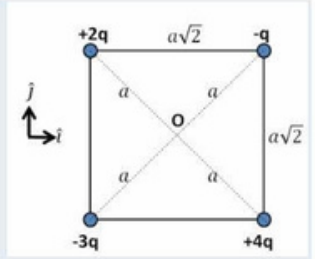
☐ F F F T T

☐ T F T F F

☐ T T T F F

☐ F F T T T

17



In the figure charges have been placed at the corner of a square where the distance from charges to the center O point is a . What is the electric potential at the center of the square? (4 Puan)

$$V = \frac{kq}{a}$$

☐ 2V

☐ 11V

☐ $2\sqrt{2}V$

☐ 10V

☐ V

18

A conducting wire has radius $r = 0.3$ mm and is 2.7 m long and it carries a current of 1 A. What's the resistance of the wire? (4 Puan)

$$\rho = 3.2 \times 10^{-8} \text{ Ohm.m}$$

☐ 0.3 Ohm

☐ 33.3 Ohm

☐ 0.27 Ohm

☐ 0.03 Ohm